Induksi Elektromagnetik



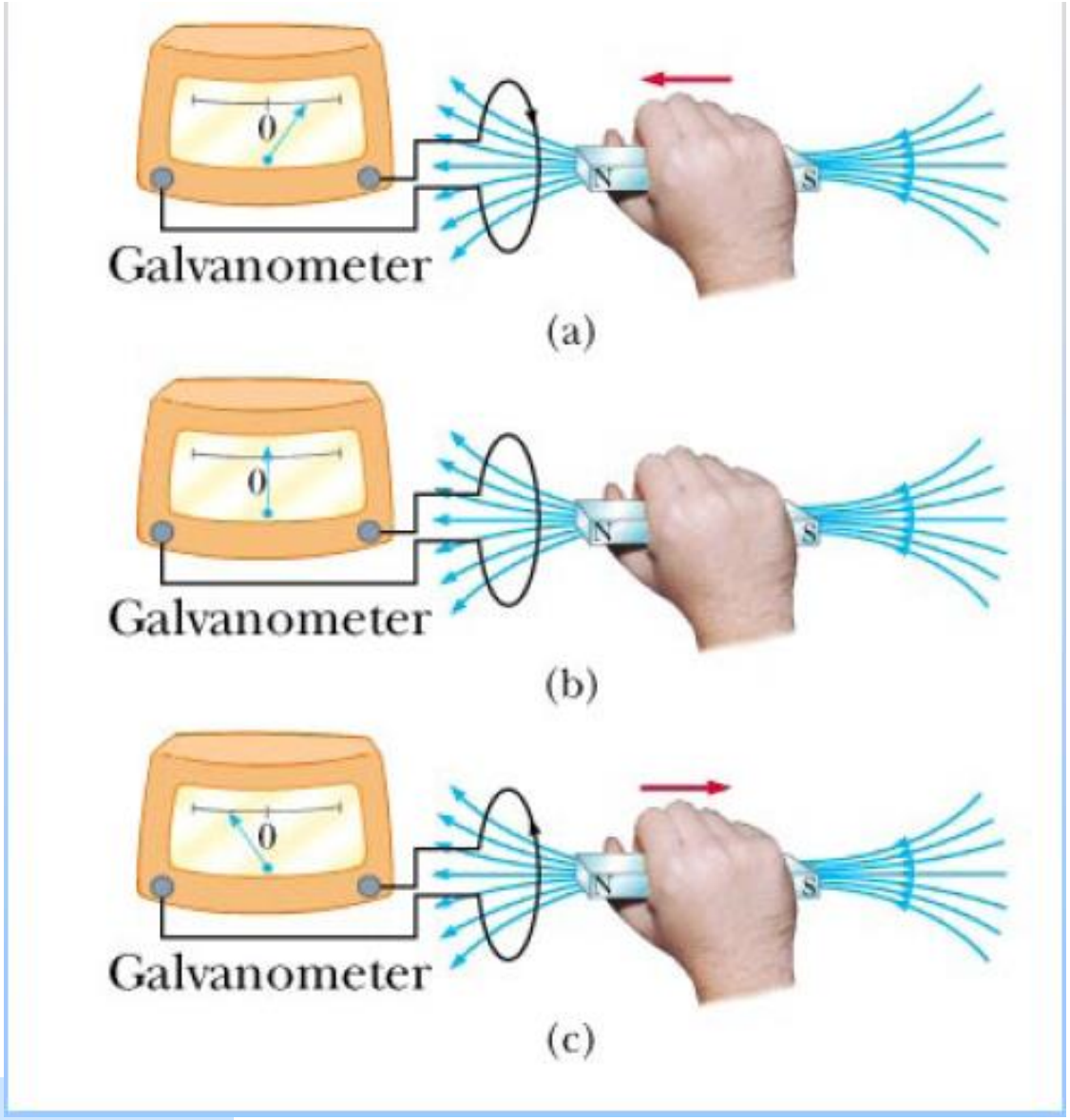
Dosen : Defrianto Pratama, S.Pd, M.Si.

Teknik Elektronika

Politeknik Negeri Bandung

Sumber bacaan Ebook: Halliday ch 30

Induksi Elektromagnetik



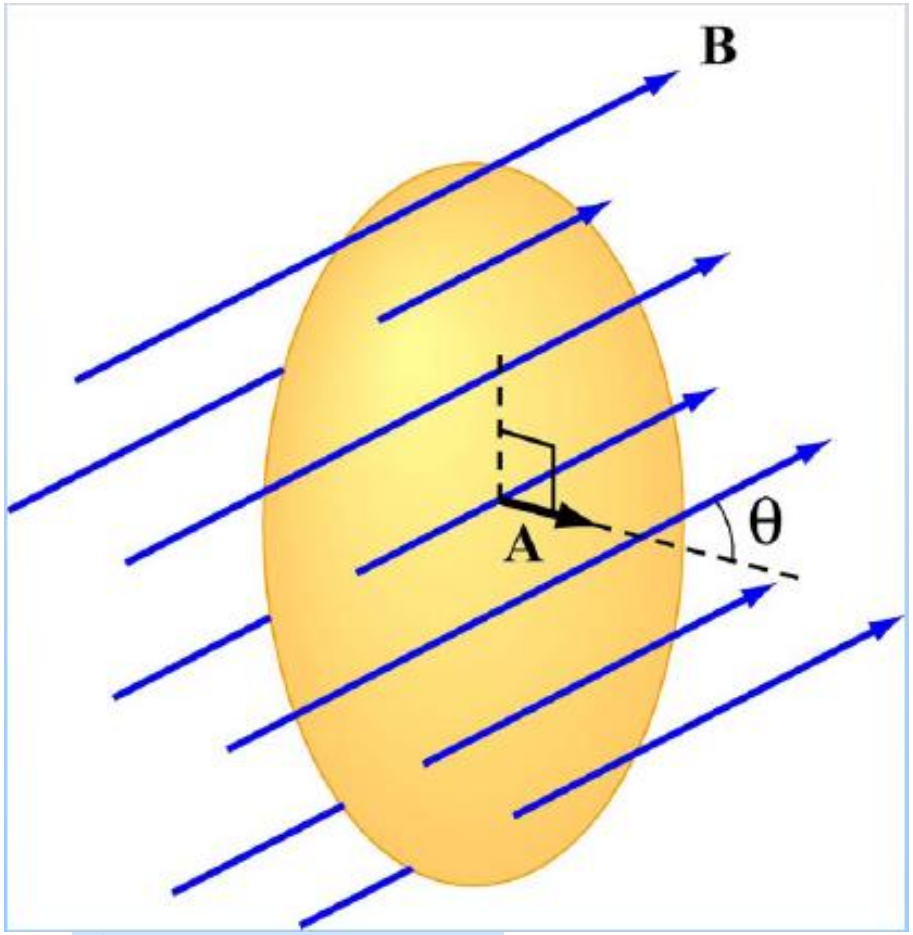
Hukum Faraday tentang Induksi

$$\mathcal{E} = -N \frac{d\Phi_B}{dt}$$

Perubahan fluks magnet menginduksi GGL

Fluks Magnet yang Menembus Loop

Analog dengan Fluks Listrik (Hukum Gauss)



(1) **B** Uniform

$$\Phi_B = B_\perp A = BA\cos\theta = \vec{\mathbf{B}} \cdot \vec{\mathbf{A}}$$

(2) **B** Non-Uniform

$$\Phi_B = \int_S \vec{\mathbf{B}} \cdot d\vec{\mathbf{A}}$$

Animasi 8.2

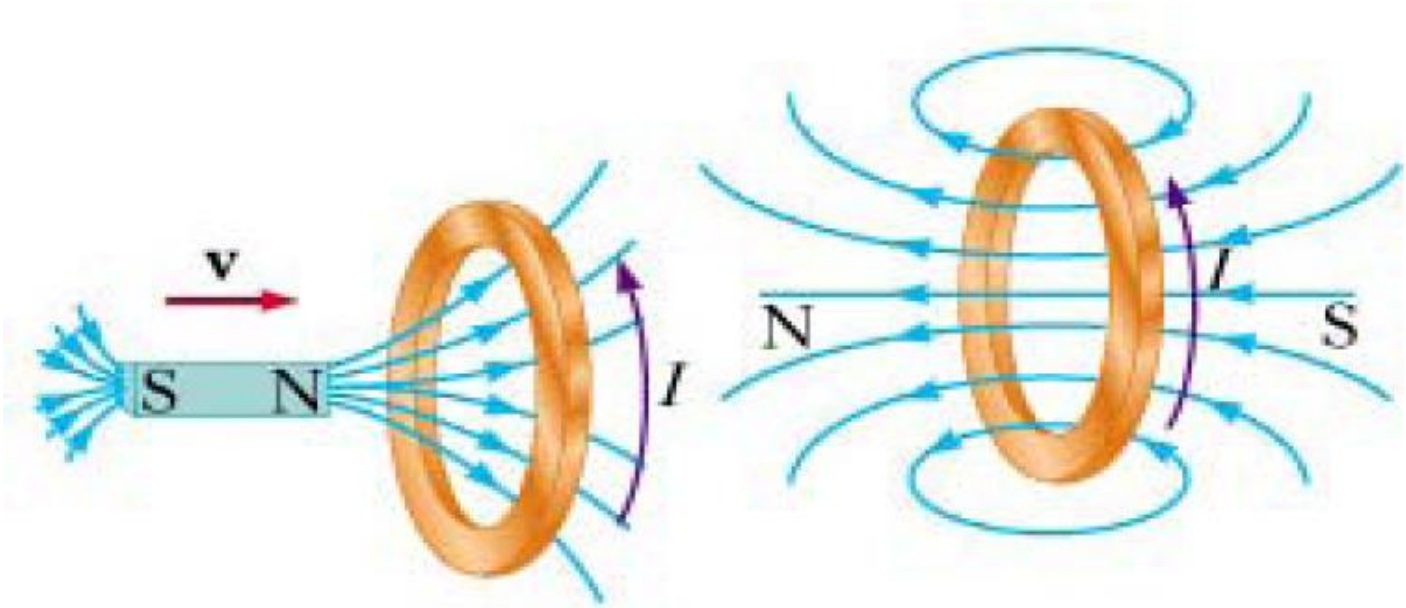
Hukum Faraday tentang Induksi

$$\mathcal{E} = -N \frac{d\Phi_B}{dt}$$

Perubahan fluks magnet menginduksi GGL

Tanda Negatif? Hukum Lenz

GGL Induksi yang muncul berarah melawan perubahan fluks yang menyebabkannya



Hukum Lenz: Arus induksi menghasilkan medan magnet yang melawan perubahan fluks magnet yang menghasilkan arus induksi tersebut

Cara untuk Menginduksi GGL

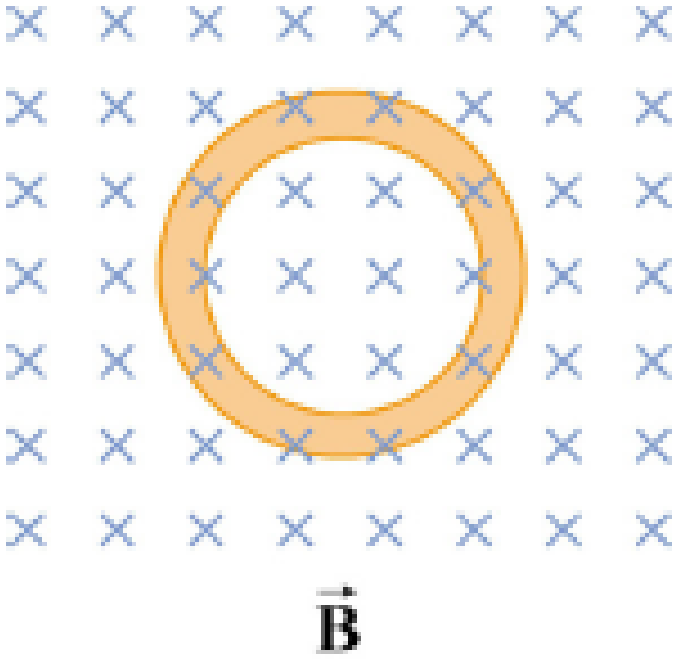
$$\mathcal{E} = -N\frac{d}{dt}\left(BA\cos\theta\right)$$

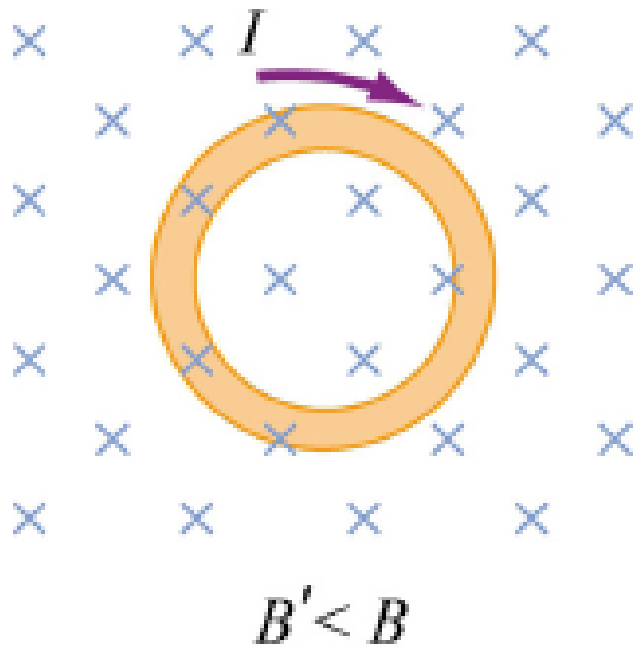
Kuantitas yang berubah terhadap waktu:

1. Besar B
2. Luas A yang dilingkupi loop
3. Sudut θ antara B dan normal loop

Cara untuk Menginduksi GGL

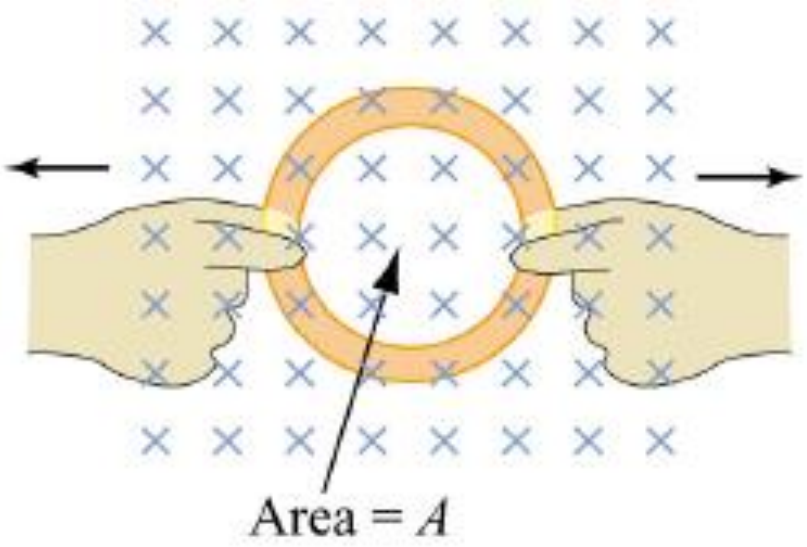
1. Besar B berubah terhadap waktu:

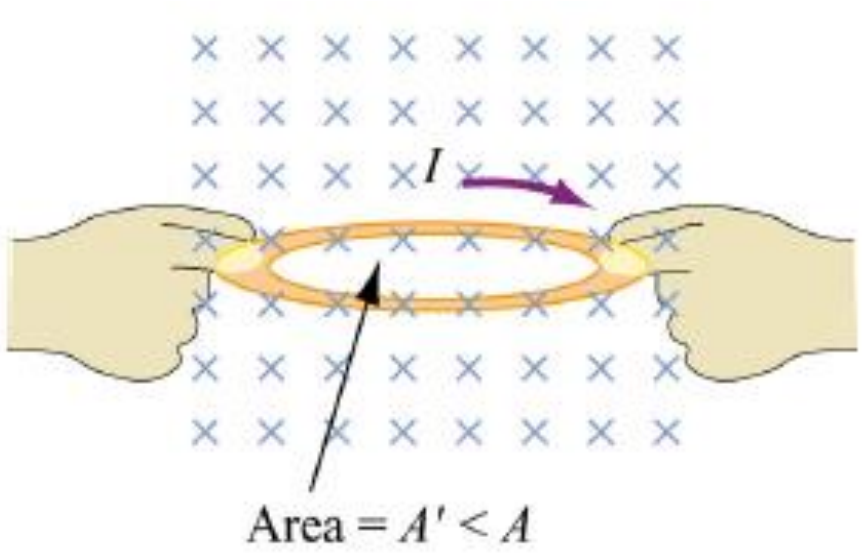




Cara untuk Menginduksi GGL

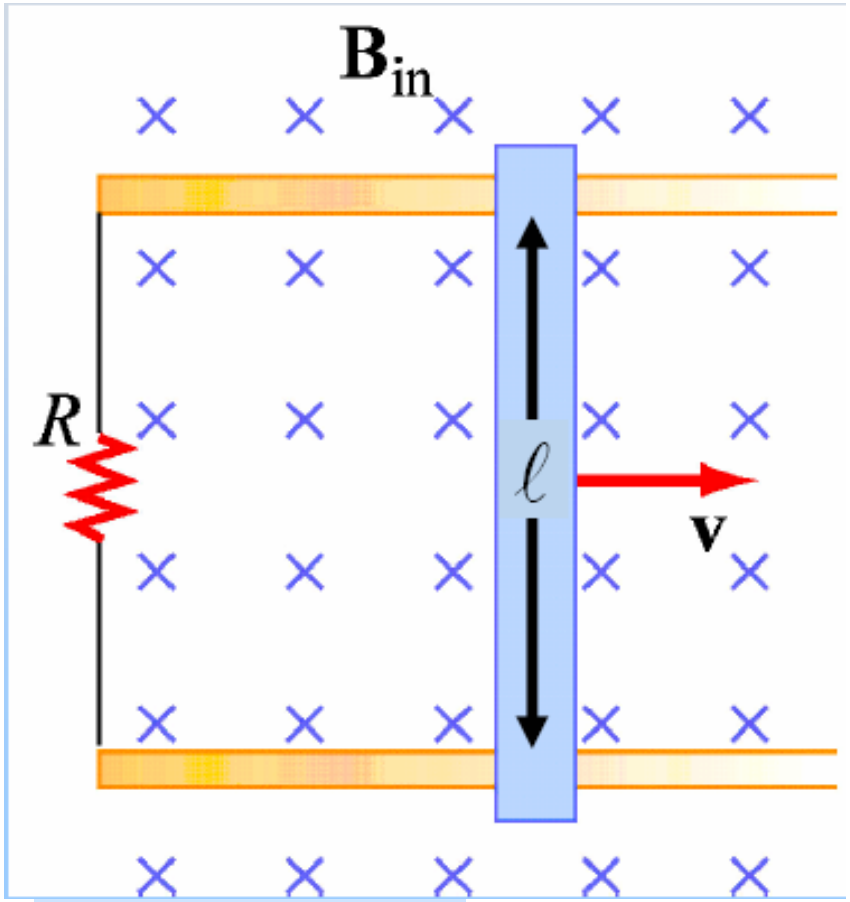
2. Luas A yang dilingkupi loop berubah terhadap waktu:





Problem: Perubahan Luas

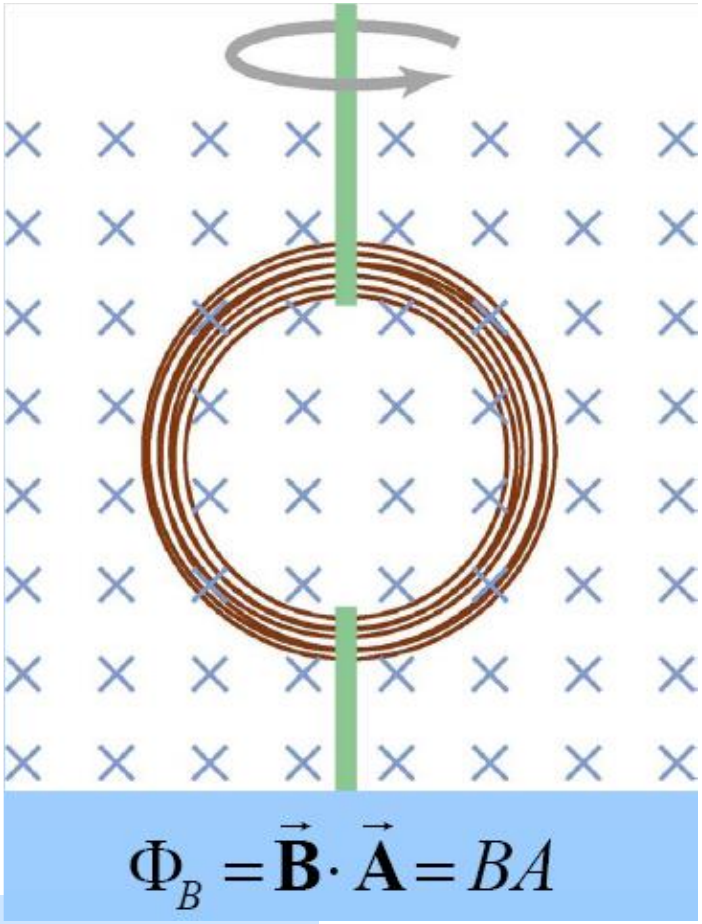
Batang konduktor ditarik sepanjang dua rel konduktor dalam daerah bermedan magnet uniform B dengan kecepatan konstan **v**



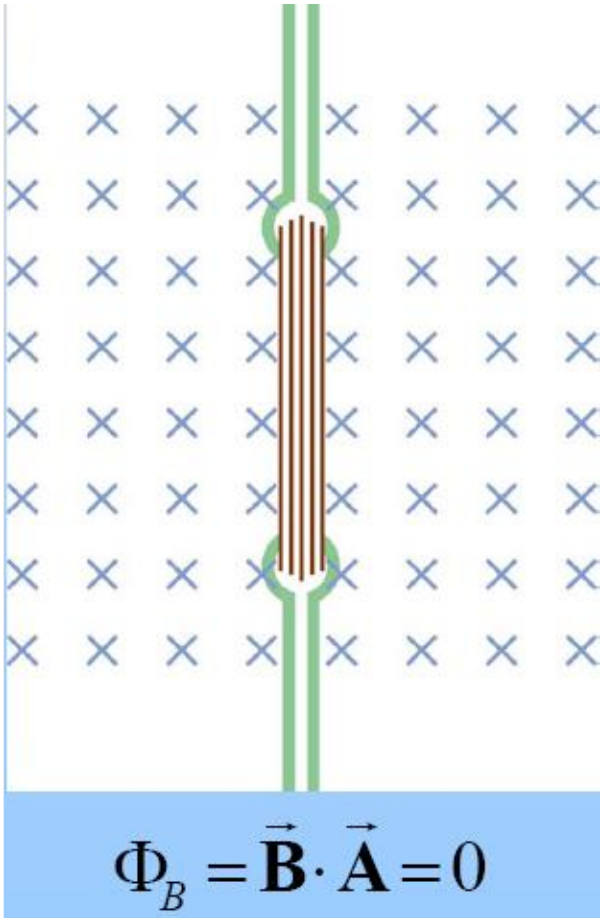
1. Arah arus induksi?
2. Arah resultan gaya?
3. Besar GGL?
4. Besar arus?
5. Daya eksternal yang harus disuplai agar batang bergerak dengan kecepatan konstan **v**?

Cara untuk Menginduksi GGL

3. Sudut θ antara B dan normal loop berubah terhadap waktu:



$$\Phi_B = \vec{\mathbf{B}} \cdot \vec{\mathbf{A}} = BA$$



$$\Phi_B = \vec{\mathbf{B}} \cdot \vec{\mathbf{A}} = 0$$

Hukum Faraday

Persamaan Maxwell

Penghasil Medan Listrik:

$$\oiint_S \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{Q_{in}}{\varepsilon_0}$$

Hukum Gauss

$$\oint_C \vec{\mathbf{E}} \cdot d\vec{\mathbf{s}} = -\frac{d\Phi_B}{dt}$$

Hukum Faraday

Penghasil Medan Magnet:

$$\oiint_S \vec{\mathbf{B}} \cdot d\vec{\mathbf{A}} = 0$$

Hukum Gauss Magnet

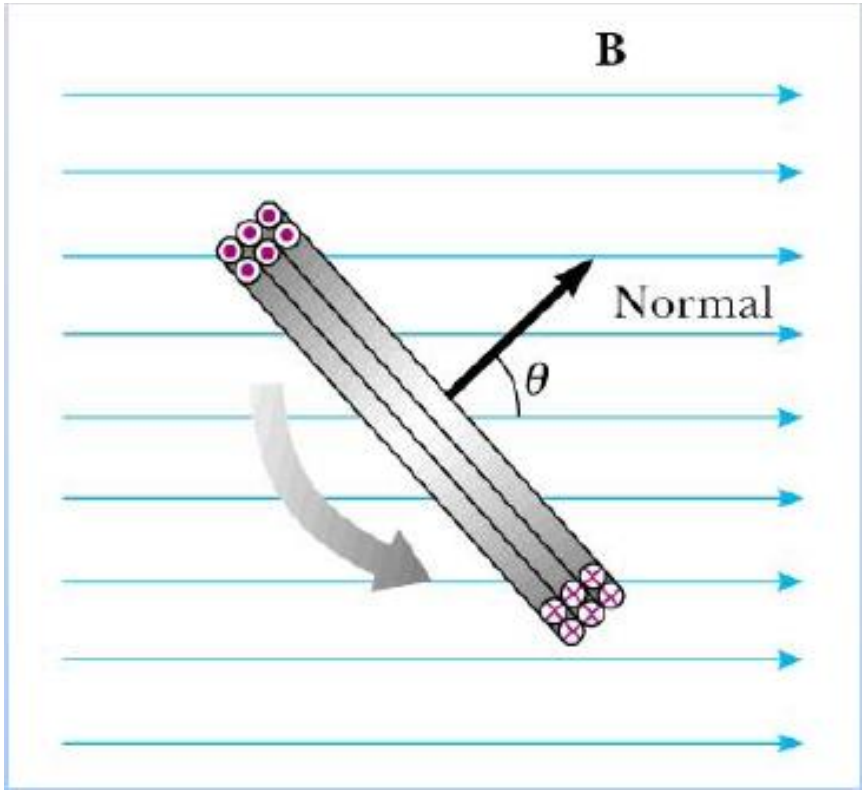
$$\oint_C \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} = \mu_0 I_{enc}$$

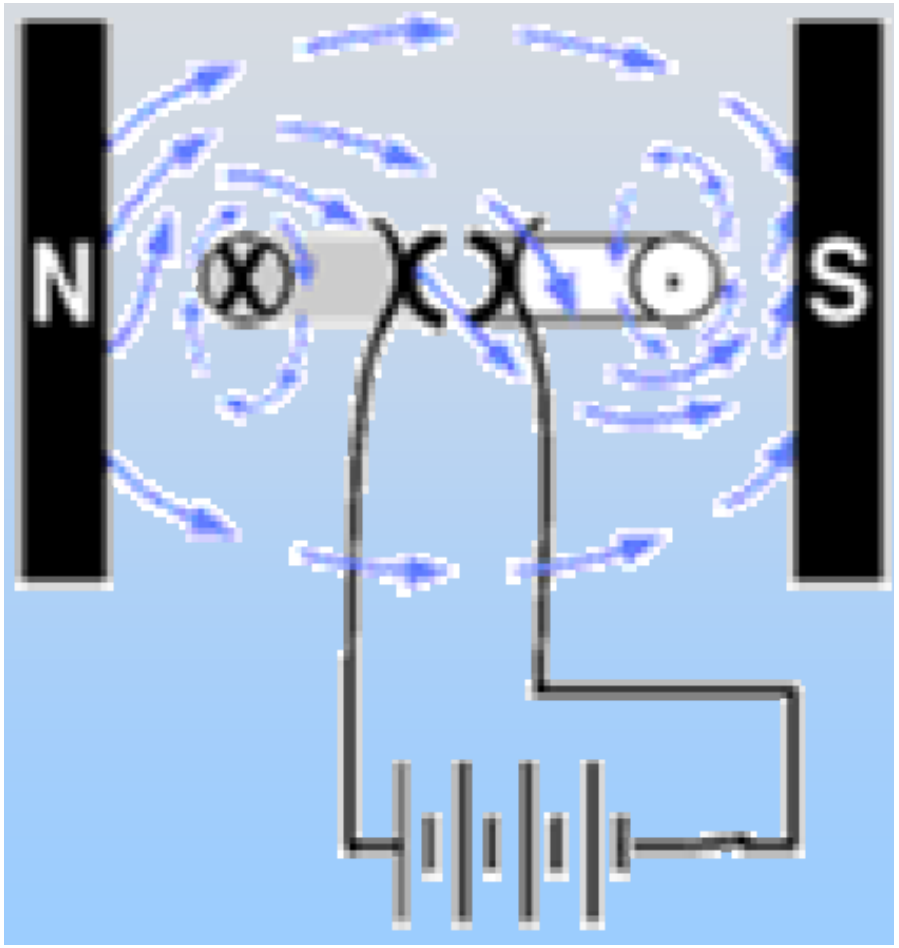
Hukum Ampere

Teknologi

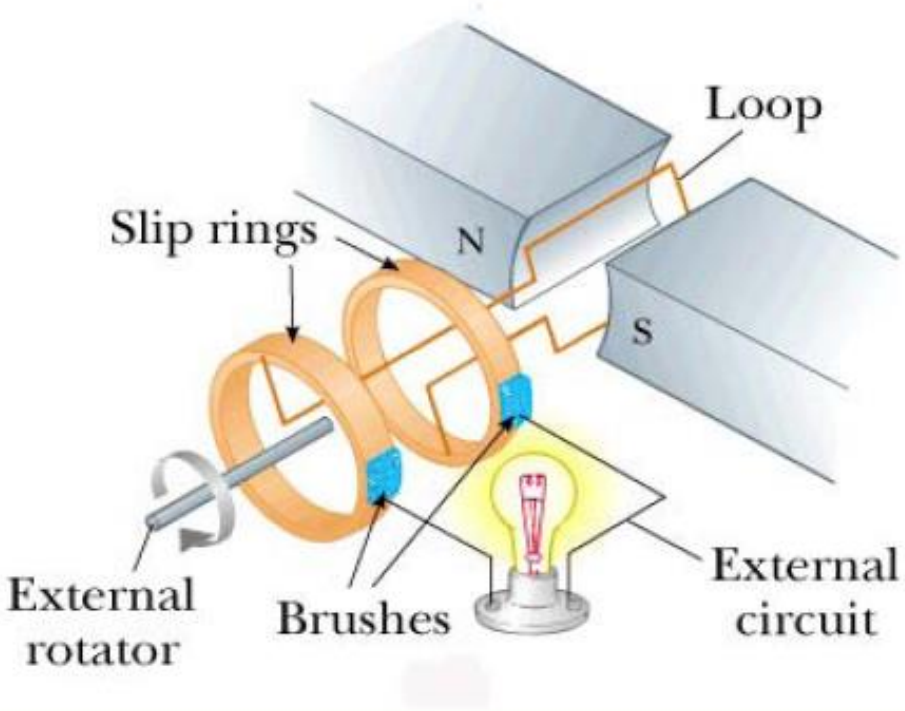
Banyak Aplikasi dari
Hukum Faraday

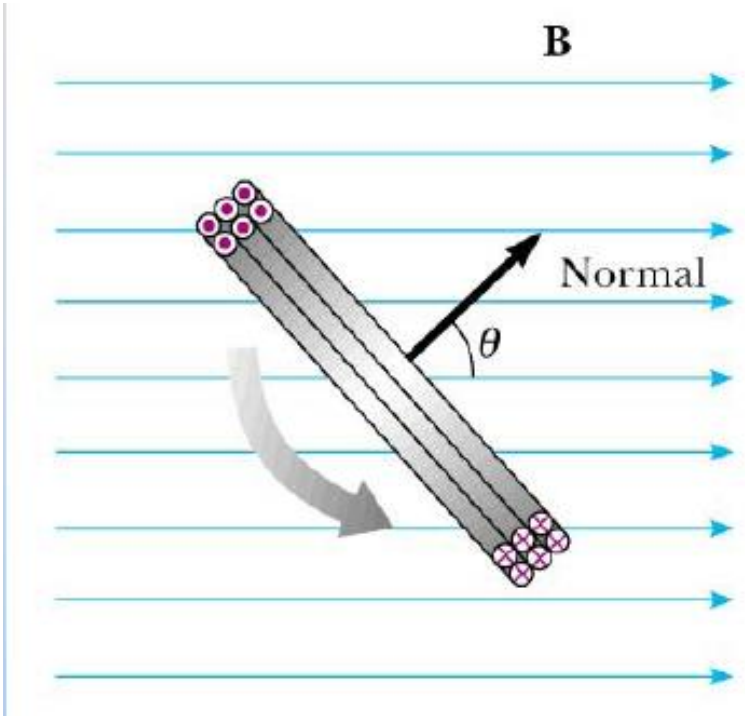
DC Motor (magnetostatik)





Motor & Generator



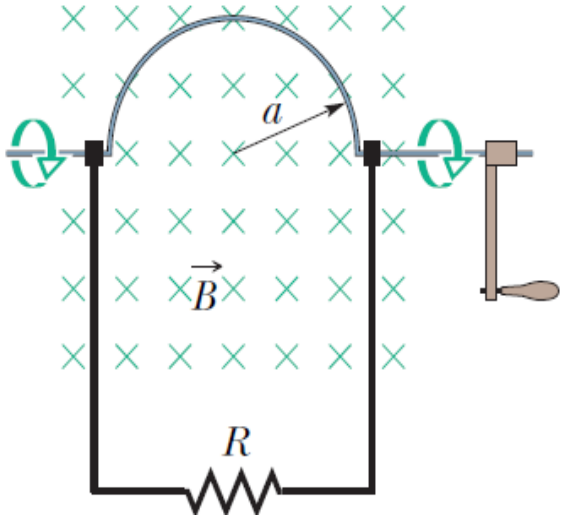


$$\Phi_B = BA\cos\theta = BA\cos\omega t$$

$$\mathcal{E} = -N\frac{d\Phi_B}{dt} = -NAB\frac{d}{dt}(\cos\omega t) = NAB\omega\sin\omega t$$

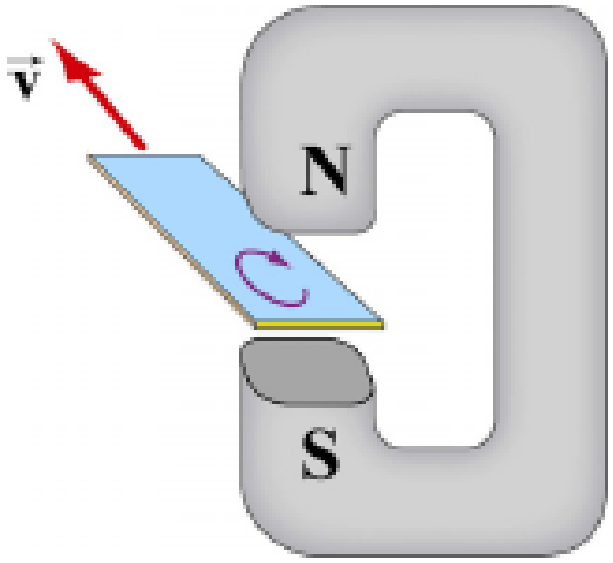
Contoh

••21 In Fig. 30-44, a stiff wire bent into a semicircle of radius $a = 2.0$ cm is rotated at constant angular speed 40 rev/s in a uniform 20 mT magnetic field. What are the (a) frequency and (b) amplitude of the emf induced in the loop?



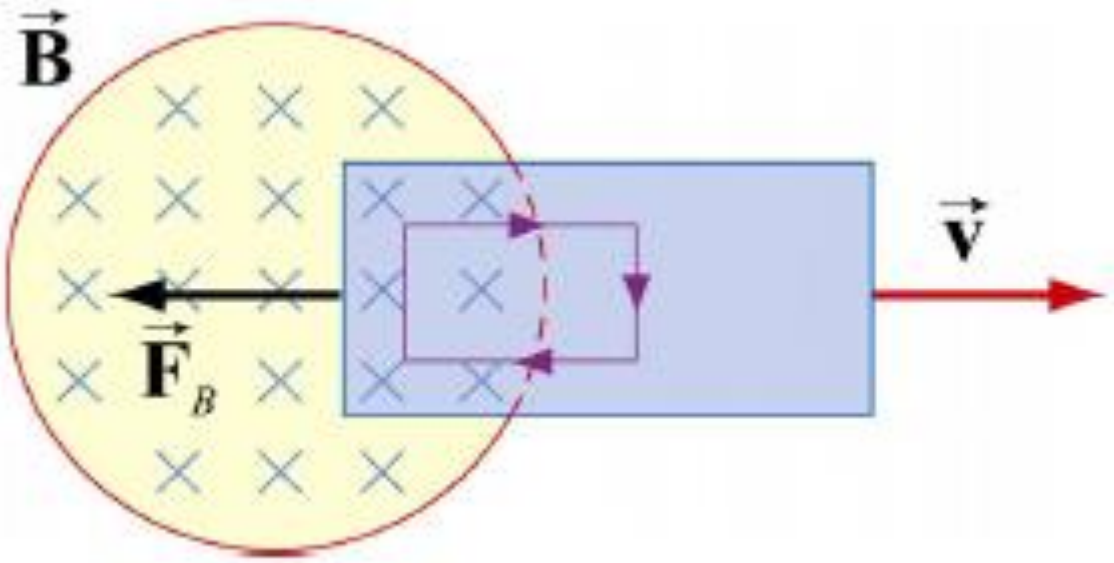
Arus Pusar (Eddy Current)

- Ketika sebuah konduktor bergerak melalui medan magnet, arus terinduksi sebagai hasil dari perubahan fluks magnet.
- Jika konduktor yang digunakan bukan berbentuk loop, arus tetap dapat terinduksi.
- Arus induksi yang muncul akan melingkar dan dinamakan *arus eddy*.



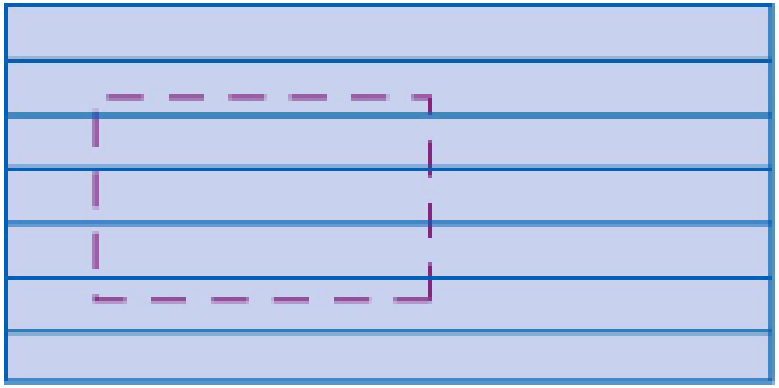
Arus Pusar (Eddy Current)

Arus eddy yang terinduksi juga menghasilkan gaya magnet yang melawan gerak, sehingga konduktor mengalami kesulitan untuk bergerak melewati medan magnet

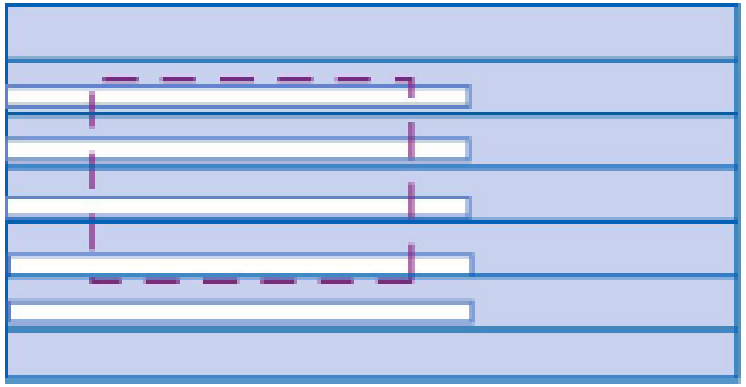


Arus Pusar (Eddy Current)

- Konduktor memiliki hambatan R
- Arus induksi muncul maka muncul daya disipasi
- Daya disipasi dapat direduksi dengan cara:



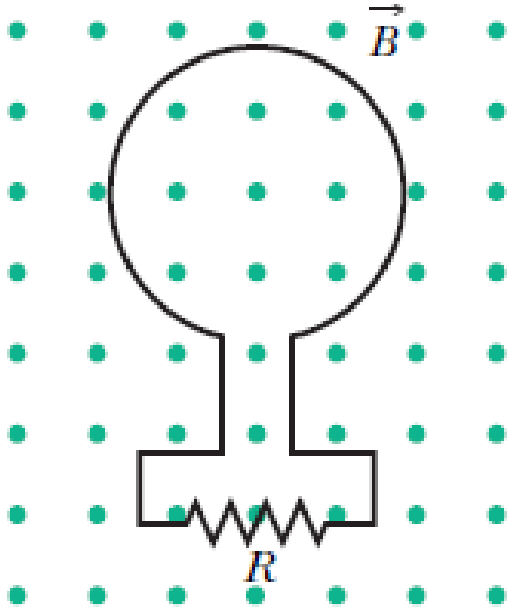
Lempengan konduktor dibungkus isolator



Lempengan konduktor dipotong-potong

Tugas No 1

•7 In Fig. 30-36, the magnetic flux through the loop increases according to the relation $\Phi_B = 6.0t^2 + 7.0t$, where Φ_B is in milliwebers and t is in seconds. (a) What is the magnitude of the emf induced in the loop when $t = 2.0$ s? (b) Is the direction of the current through R to the right or left?



Tugas No 2

••34 In Fig. 30-53, a long rectangular conducting loop, of width L, resistance R, and mass m, is hung in a horizontal, uniform magnetic field $\vec{B}$ that is directed into the page and that exists only above line aa. The loop is then dropped; during its fall, it accelerates until it reaches a certain terminal speed v_t. Ignoring air drag, find an expression for v_t.

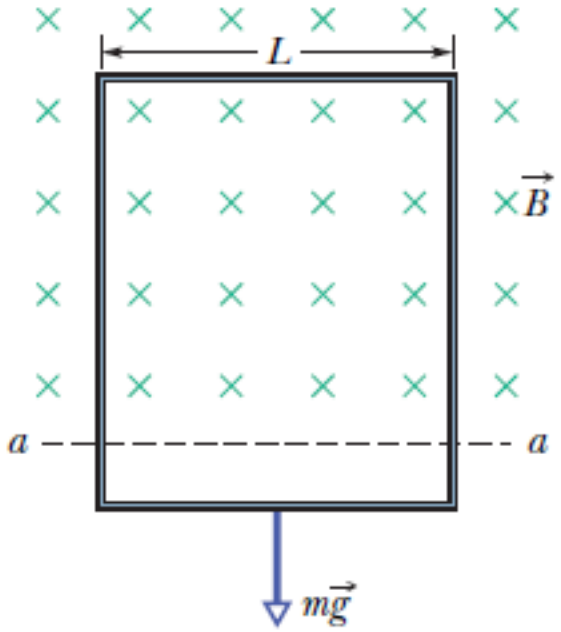


Fig. 30-53 Problem 34.